The Planetary Science Journal, 7:18 (13pp), 2026 January https://doi.org/10.3847/PSJ/ae2b57
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# Remote Compositional Analyses of Space-weathered Lunar Maria

Ji-In Jung[1], Mathieu G. A. Lapôtre[2], Ralph E. Milliken[3], and Sarah E. Minson[4]
[1] Department of Geophysics, Stanford University, Stanford, CA 94305, USA; jiinjung@stanford.edu
[2] Department of Earth and Planetary Sciences, Stanford University, Stanford, CA 94305, USA
[3] Department of Earth, Environmental and Planetary Sciences, Brown University, Providence, RI 02912, USA
[4] U.S. Geological Survey Earthquake Science Center, Menlo Park, CA 94025, USA
*Received 2025 March 15; revised 2025 November 20; accepted 2025 December 4; published 2026 January 22*

## Abstract

Visible-to-shortwave infrared (VSWIR) reflectance spectroscopy has revolutionized our understanding of planetary surface compositions. However, space-weathering processes on airless bodies complicate quantitative compositional analyses. Here, we present a framework to isolate the signatures of space weathering in VSWIR spectra of lunar maria by leveraging radiative transfer modeling under the assumptions that (i) a space-weathered target can be expressed as a mixture of fresh and fully space-weathered components and (ii) remaining signatures can be modeled by including agglutinates as an end-member component. We first validate this approach against laboratory spectra of space-weathered Apollo mare soils of known mineral compositions using a probabilistic Markov Chain Monte Carlo implementation of the Hapke radiative transfer model. Second, we illustrate how this approach can be applied to orbital Moon Mineralogy Mapper data. The proposed space-weathering correction workflow for lunar maria could be expanded to other lunar lithologies and applied to existing and future data sets.

*Unified Astronomy Thesaurus concepts:* Lunar maria (961); Infrared spectroscopy (2285); Remote sensing (2191); Space weather (2037); Spectroscopy (1558); Lunar mineralogy (962)

## 1. Introduction

Applications of visible-to-shortwave infrared (VSWIR) reflectance spectroscopy in multiband and hyperspectral data sets have provided abundant information on the compositions of planetary surfaces, including interpretations of primary and secondary mineralogy. Numerous studies have utilized VSWIR spectroscopy for basic qualitative identification of various minerals based on their distinctive electronic and vibrational absorptions, including for the Moon, Mars, Ceres, and near-Earth asteroids (C. M. Pieters 1986; J.-P. Bibring et al. 2005; R. E. Arvidson et al. 2014; B. L. Berg et al. 2016; D. C. Reuter et al. 2018; L. Riu et al. 2021). Less commonly, some studies have attempted to quantify the abundance of various mineral phases on planetary bodies using different radiative transfer models (RTMs; B. Hapke 1981; Y. Shkuratov et al. 1999; S. J. Lawrence & P. G. Lucey 2007; F. Poulet et al. 2008).

Under classic RTM approaches, a single set of minerals is inferred from the best-fit modeled spectrum to the data (J. T. Cahill & P. G. Lucey 2007; B. W. Denevi et al. 2008; J. T. S. Cahill et al. 2010; S. Li & L. Li 2011), which might not be representative of actual mineralogy due to the non-unique nature of the inverse problem (M. G. A. Lapôtre et al. 2017a). In contrast, probabilistic implementations of such models recognize the non-uniqueness of spectral unmixing and instead output ranges of possible mineral assemblages (M. G. A. Lapôtre et al. 2017a, 2017b; H. Kurokawa et al. 2020; I. Mishra et al. 2021; C. Pilorget & J. Fernando 2021; O. King et al. 2022). Other approaches to modeling VSWIR reflectance spectra include the modified Gaussian method (J. M. Sunshine et al. 1990; J. M. Sunshine & C. M. Pieters 1993; S. K. Noble et al. 2006; H. Clénet et al. 2011; P. J. Isaacson et al. 2011) and shallow machine learning techniques (S. Li et al. 2012; M. Hareyama et al. 2019; G. R. Kodikara & L. J. McHenry 2020). Retrieving quantitative estimates of mineralogy from reflectance spectra of planetary surfaces that have been significantly modified by space weathering—such as the Moon—has received even less attention.

Building on early telescopic observations, reflectance data acquired from lunar orbit by the Clementine and Lunar Prospector missions facilitated the identification of spectrally and presumably mineralogically distinct regions at ultraviolet-to-visible wavelengths (B. L. Jolliff et al. 2000; C. M. Pieters et al. 2001; P. G. Lucey 2004). More recently, the Kaguya Multiband Imager (9 bands; M. Ohtake et al. 2008), Kaguya Spectral Profiler (296 spectral channels; T. Matsunaga et al. 2008), and Chandryaan1 Moon Mineralogy Mapper (M3; 86 spectral channels in global mode and 260 in targeted mode; J. N. Goswami & M. Annadurai 2008; C. M. Pieters et al. 2009; R. O. Green et al. 2011) instruments have allowed scientists to map the distribution of water, constrain volcanic stratigraphy, and detect small outcrops of olivine-rich, Fe-plagioclase-rich, and spinel-bearing zones (C. M. Pieters et al. 2009; C. Pieters et al. 2014; M. Hareyama et al. 2019; M. Lemelin et al. 2019).

However, quantifying mineral abundances on the Moon has remained challenging due to the strong spectral modifications imparted by space-weathering processes (L. A. Taylor et al. 2000; D. Trang & P. G. Lucey 2019; Figure 1). These include micrometeorite bombardment and interactions with solar winds and result in the formation of (1) impact-vitrified glass in agglutinates (aggregates of soil particles; B. Hapke 2001; C. N. Yasanayake et al. 2024) and (2) nanophase (and microphase) metallic iron ($npFe^0$) on the rims (or inside) of mineral grains (W. Cassidy & B. Hapke 1975; C. M. Pieters & S. K. Noble 2016). These phenomena tend to decrease the overall albedo, induce spectral reddening (a positive slope with

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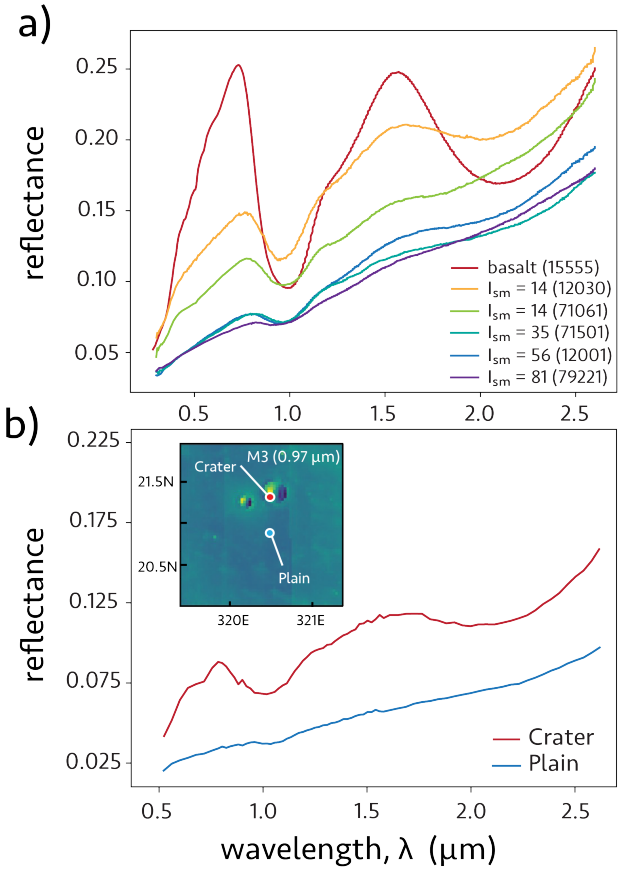



**Figure 1.** (a) Laboratory reflectance spectra of pulverized fresh basalt (15555) and lunar soils with various degrees of soil maturities (12001, 12030, 71061, 71501, and 79221), as indicated by their soil maturity ratio ($I_{sm} = I_s/\mathrm{FeO}$), where $I_s = (\Delta \mathrm{H})^2 A$, with $\Delta$H, the ferromagnetic resonance (FMR) line width, and $A$, its amplitude (R. V. Morris 1978). (b) Thermally corrected M3 reflectance spectra of freshly exposed rock inside the simple crater located near the Aristarchus crater (red) and surrounding space-weathered regolith (blue); M3 data processed according to S. Li & R. E. Milliken (2016). Inset map shows the location of those spectra (overlaid on 970 nm reflectance base map).

increasing wavelength), and weaken absorption features that are diagnostic of mineralogy (Figure 1).

Many studies have attempted to parameterize the degree of space weathering across the lunar surface or to use certain magnetic properties to estimate the amount of npFe$^0$ (which increases with the degree of space weathering; B. Hapke 2001; S. Li & L. Li 2011; P. G. Lucey & M. A. Riner 2011; D. Trang & P. G. Lucey 2019) through empirical parameters such as optical maturity (P. G. Lucey et al. 2000a, 2000b) or the soil maturity index ($I_{sm}$) (R. V. Morris 1978). Studies have also explored methods to minimize or reduce spectral effects of space weathering, such as through spectral continuum removal (C. M. Pieters et al. 2019). However, current spectral methods are not capable of removing or accounting for the spectral signatures of space weathering in a way that allows for quantitative estimates of modal mineralogy of lunar regolith. One potential exception is the radiative transfer modeling of B. Hapke (2001). Grounded in scattering theory, this approach requires (a) the assignment of a preferred distribution of npFe in the grains (e.g., homogenous throughout the grain versus concentrated in rims) and (b) the treatment of the abundance and size of npFe particles separately for each mineral phase in the mixture, doubling the number of unknowns in the inverse problem. Furthermore, the simultaneous presence of both nanophase and microphase iron, which follow different scattering regimes in the VSWIR range, is not taken into account in the model of B. Hapke (2001), further challenging its applicability to lunar regolith. Here, we present a new, semiempirical framework to quantify modal mineralogy of space-weathered regolith in lunar maria.

## 2. Approach

### 2.1. Step 1: Semiempirical Band Depth Correction

Using laboratory spectra of Apollo soil samples, we devised a workflow to remove the spectral effects of space weathering from VSWIR spectra of lunar materials (e.g., lunar regolith) and estimate reflectance spectra of the "fresh" (unweathered) material from which quantitative estimates of mineralogy can be derived. We refer to the first step as a "band depth correction" though it also affects spectral slopes. Qualitatively, this step consists of scaling the band depth of a major absorption in spectra affected by space weathering (specifically, by npFe$^0$) to restore a fresh, unaltered appearance to the spectrum, with corrected deeper absorption features.

Following the Hapke RTM for intimate mixtures (B. Hapke 1981), we assume that the single scattering albedo (SSA) of a target with some degree of space weathering, $\omega_{\mathrm{mix}}$, can be modeled as a linear mixture of a "depth-corrected" SSA spectrum ($\omega_{\mathrm{dc}}$) and that of a fully space-weathered target ($\omega_{\mathrm{sw}}$). This linear combination is weighted by the proportions of the depth-corrected ($\theta_{\mathrm{dc}}$) and space-weathered ($\theta_{\mathrm{sw}}$) components (with $\theta_{\mathrm{dc}} + \theta_{\mathrm{sw}} = 1$), i.e.,

$$\omega_{\mathrm{mix}} = \theta_{\mathrm{dc}}\omega_{\mathrm{dc}} + \theta_{\mathrm{sw}}\omega_{\mathrm{sw}}, \quad (1)$$

such that

$$\omega_{\mathrm{dc}} = \frac{\omega_{\mathrm{mix}} - \theta_{\mathrm{sw}}\omega_{\mathrm{sw}}}{1 - \theta_{\mathrm{sw}}}. \quad (2)$$

In turn, the degree of space weathering, $\theta_{\mathrm{sw}}$ is assumed to be proportional to the soil maturity index ($\theta_{\mathrm{sw}} = I_{\mathrm{sm}}/100$). The soil maturity index serves as a semiquantitative measure to classify lunar soil maturity and is typically measured as the ratio of ferromagnetic resonance (FMR) surface exposure index ($I_s$) to total iron content, i.e., $I_{\mathrm{sm}} = I_s/\mathrm{FeO}$. Ratios of 0–29, 30–59, and greater than 60 indicate immaturity, submaturity, and maturity, respectively (R. V. Morris 1978).

To establish a relationship between $\theta_{\mathrm{sw}}$ and band depth in lunar maria, we use data from the Lunar Soil Characterization Consortium (LSCC) database (C. M. Pieters et al. 2000; L. A. Taylor et al. 2001). The LSCC database consists of a comprehensive spectral characterization of lunar mare soils selected to represent a range of maturity states and grain sizes conducted by a collaborative group of lunar sample and remote sensing scientists. It includes detailed analyses of chemical and mineralogical abundances performed through sieving of pristine lunar soils, imaging analysis, electron microprobe chemical analysis, FMR analysis, and spectral reflectance measurements. Their reflectance spectra, spanning 0.3–2.6 $\mu$m, were acquired at a 5 nm sampling resolution using the RELAB bidirectional spectrometer. We find that $I_{\mathrm{sm}}$ correlates with the ratio of the ~1000 nm Fe$^{2+}$ absorption depth to that of a fully

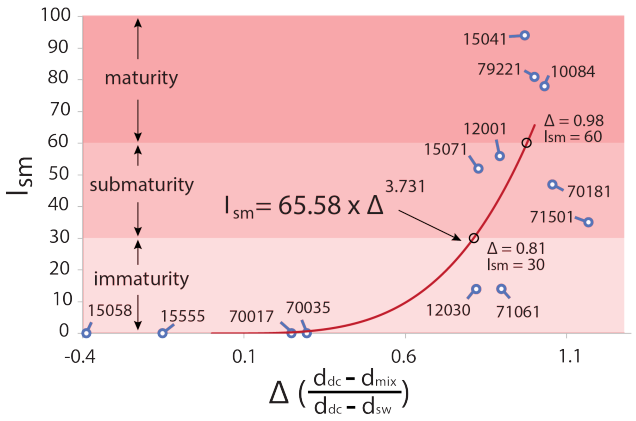



**Figure 2.** Relationship between the 970 nm band depth and soil maturity, $I_{sm} = I_s/\mathrm{FeO}$, as observed in Apollo soil samples (R. V. Morris 1978). The red line indicates the fitted relationship used to link $\Delta$ to $I_{sm}$. Spectral data for original mare basalts (P. J. Isaacson et al. 2011) and Lunar Soil Characterization Consortium (LSCC) samples (C. M. Pieters et al. 2000; L. A. Taylor et al. 2001) are available at https://sites.brown.edu/relab/lscc/ and https://pds-speclib.rsl.wustl.edu/.

space-weathered surface through

$$\theta_{sw} = \frac{I_{sm}}{100} \approx 0.6568 \times \Delta^{3.731}, \quad (3)$$

where $\Delta$ is defined as

$$\Delta = \frac{d_{dc}(\lambda = 970\ \mathrm{nm}) - d_{mix}(\lambda = 970\ \mathrm{nm})}{d_{dc}(\lambda = 970\ \mathrm{nm}) - d_{sw}(\lambda = 970\ \mathrm{nm})}, \quad (4)$$

with $d(\lambda)$ representing the continuum-removed SSA band depth value at the wavelength $\lambda$, and subscripts "dc," "mix," and "sw" refer to "depth corrected," "mixture," and "space weathered," respectively (Figure 2). Equation (3) highlights the nonlinearity of band depth with the degree of space weathering. A reference wavelength of $\lambda =$ 970 nm was adopted because it is the approximate position of a prominent and common $Fe^{2+}$ absorption feature observed in reflectance spectra of the Moon, the strength of which is highly correlated with the degree of space weathering (Figure 2; Appendix A; Figure A1).

Although the soil maturity index ($I_{sm}$) of LSCC samples was directly measured and reported (R. V. Morris 1978), its value is not readily available for orbiter-based remote-sensing observations. Thus, using Equation (3) allows us to ensure applicability of the workflow to orbiter-based spectral observations. In the following, the adopted SSA spectrum for a fully weathered target, $\omega_{sw}$, varies depending on whether Apollo samples or orbiter-based spectra are considered. For Apollo soils, the SSA spectrum of a fully space-weathered end-member, $\omega_{sw}$, was taken as the average spectrum of samples 79221 ($I_{sm} = 81$), 15041 ($I_{sm} = 94$), and 10084 ($I_{sm} = 78$) (Figure 3). For remote-sensing data, it is taken as the average spectrum of the most space-weathered pixels near the target region of interest (ROI).

### 2.2. Step 2: Spectral Modeling

The first step in the workflow yields a "depth-corrected" spectrum, $\omega_{dc}$, which is primarily intended to account for the influence of $npFe^0$ on absorption strength. However, it does not fully account for how space-weathering influences spectral slope, and it does not yet account for other spectral properties associated with agglutinates, an important product of lunar space weathering. Unlike components such as pyroxene, olivine, or Fe-rich glass, agglutinates have almost no absorption features and can be spectrally dominated by $npFe^0$ (i.e., exhibit a red-sloped component in their spectrum). The next step consists of estimating the residual signatures of space weathering (including the red spectral slope) through spectral unmixing of the intermediate, "depth-corrected" spectrum, including agglutinates as a compositional end-member, and ultimately removing them by forward modeling a final corrected spectrum, $\omega_f$, by excluding the contribution of the agglutinate component.

Due to the non-unique nature of spectral unmixing inversions, we employed a Bayesian implementation of the Hapke model, following M. G. A. Lapôtre et al. (2017a), to unmix "depth-corrected" spectra ($\omega_{dc}$) and retrieve modal estimates of dominant phases and corresponding particle sizes. The inverse problem is solved using the cascading adaptive transitional metropolis in parallel (CATMIP) algorithm (S. E. Minson et al. 2013, 2014), a parallel Markov Chain Monte Carlo (MCMC) algorithm (based on the Metropolis–Hastings algorithm) that efficiently samples high-dimensional spaces. This step provides quantitative estimates of modal mineralogy from spectral data.

Prior distributions for mineral abundances were drawn from Dirichlet distributions, ensuring that the sum of all abundances is equal to 100%. The grain size of each end-member was parameterized as a separate variable drawn from uniform prior distributions, ranging from ~0 to 990 $\mu$m in the initial setup (Appendix B). Nine phases commonly found in lunar mare basalts were selected as compositional end-members: olivine (fayalitic olivine, Fo5-10), pigeonite, augite, plagioclase, ilmenite, glass of pyroxene-like composition, glass of plagioclase-like composition, Apollo orange glass, and agglutinates (Appendix C; Figure C1; Table C1). Therefore, a total of 18 parameters were considered. The glass with pyroxene-like and plagioclase-like compositions refers to synthetic amorphous glasses composed of pyroxene, plagioclase, and ilmenite in weight percentages of 66.7%/26.7%/6.7% and 37.0%/59.3%/3.7%, respectively (E. A. Cloutis et al. 1990). As this MCMC algorithm takes the optical constants of each mineral as input (M. G. A. Lapôtre et al. 2017a), for each mineral end-member, we first converted its reflectance spectra to SSA (Appendix D); the SSA values were subsequently converted to optical constants (Appendix E).

The MCMC algorithm was run with a Markov chain length of $10^6$ and outputs posterior distributions for all 18 parameters (a modal abundance vector and a grain size vector, each with nine elements), permitting evaluation of the full suite of mineral assemblages that yield a reasonable match to the data. Acceptance and rejection of sampled compositional models are decided based on the goodness of the fit between true and modeled spectra. In addition to the full posterior distributions, two specific models are considered: the maximum a posteriori, or MAP (i.e., the model determined to have the overall highest likelihood), estimation, and the mode (i.e., using the most likely value for each of the 18 parameters, which does not necessarily correspond to the MAP). From the various possible model solutions, we use the resulting mineral abundances and corresponding particle sizes to forward model the spectrum of the fresh, non-space-weathered material,

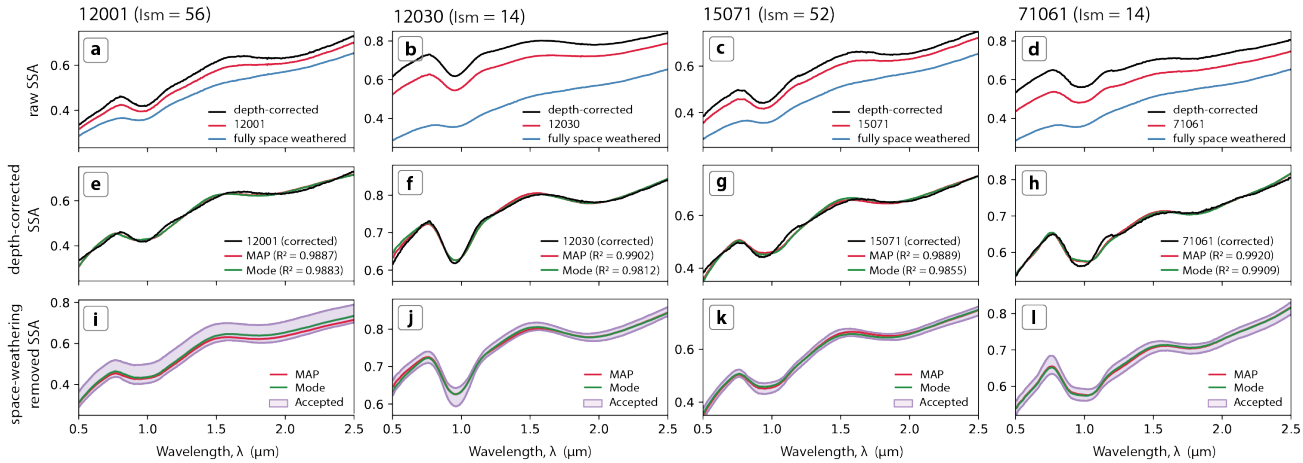



**Figure 3.** Analysis of four Lunar Soil Characterization Consortium (LSCC) mare basalt soils with varying degrees of space weathering (a), (e), (i) 12001 ($I_{sm}$= 56); (b), (f), (j) 12030 ($I_{sm}$ = 14); (c), (g), (k) 15071 ($I_{sm}$ = 52); and (d), (h), (l) 71061 ($I_{sm}$= 14). Panels (a)–(d) display the original SSA (red), the SSA of the fully space-weathered end-member (blue), and the depth-corrected SSA (black). Panels (e)–(h) show model fits to the depth-corrected SSA spectrum (black) from Bayesian spectral unmixing modeling, with MAP (red) and mode (green). Panels (i)–(l) display space-weathering-corrected SSA spectra (red = MAP, mode = green, and shading = range of accepted models across $10^6$ samples.

**Table 1**
List of LSCC Samples Used for Space-weathering Correction

| Sample ID | RELAB Code | $I_{sm}$ | $d_{dc}$ | $d_{sw}$ | $d_{mix}$ | $\Delta$ | Calculated $I_{sm}$ |
|---|---|---|---|---|---|---|---|
| 12001 | LR-CMP-010 | 56 | 0.358 | 0.120 | 0.145 | 0.893 | 43.04 |
| 12030 | LR-CMP-032 | 14 | 0.358 | 0.120 | 0.163 | 0.819 | 31.26 |
| 15071 | LR-CMP-020 | 52 | 0.358 | 0.120 | 0.161 | 0.826 | 32.13 |
| 71061 | LR-CMP-057 | 14 | 0.358 | 0.120 | 0.144 | 0.897 | 43.85 |

**Note.** $I_{sm}$: reported soil maturity index (R. V. Morris 1978); $d_{dc}$: continuum-removed SSA band depth value at the wavelength 970 nm for pulverized mare basalt samples 15058, 15555, 70017, and 70035; $d_{sw}$: continuum-removed SSA band depth value at the wavelength 970 nm for the most space-weathered samples 15041, 79221, and 10084; $d_{mix}$: continuum-removed SSA band depth value at the wavelength 970 nm for the target sample; $\Delta$: ratio of the 970 nm band depth to that of fully space-weathered samples using Equation (4); calculated $I_{sm}$: soil maturity index calculated using reported $I_{sm}$ with Equation (3).

omitting the contribution of agglutinates that was estimated from the inverse model.

## 3. Results and Interpretations

*Application to Lunar Soils*. Reflectance spectra of lunar soils derived from lunar mare basalt regions with a range of soil maturity (12001, $I_{sm}$ = 56; 12030, $I_{sm}$ = 14; 15071, $I_{sm}$ = 52; and 71061, $I_{sm}$ = 14) were used to assess the effectiveness of the method (Table 1). For all four LSCC mare basalt soil samples examined here, the first step yields intermediate, "depth-corrected" spectra with deeper absorptions, less spectral reddening, and overall higher albedos (Figure 3(a)–(d)). The rms values of accepted models for fitting the resulting $\omega_{dc}$ spectra, including the mode and MAP, consistently exceed 0.98, indicating that the depth-corrected spectra could be well reproduced from linear mixtures of SSA spectra of the nine selected compositional end-members (Figure 3(e)–(h)). Subsequent forward modeling without agglutinates yields corrected spectra that resemble those of typical lunar basalts, with pronounced 1 and 2 $\mu$m absorption bands and visibly less spectral reddening (Figure 3(i)–(l)).

Though corrected LSCC spectra look qualitatively similar to spectra of fresh lunar basalt, we further validated their quantitative robustness by comparing inverted mineral abundances with independently measured compositions or through petrological analyses reported in the Apollo lunar compendium (K. A. Cone et al. 2020). The 95% probability densities of modeled abundances show substantial overlap with laboratory measurements, demonstrating close agreement between modeled and lab-measured mineral abundances (Figure 4). In some cases, MAP estimates differ slightly from laboratory measurements, emphasizing the need for probabilistic modeling rather than relying on a single best-fit model. For olivine, differences from LSCC are generally less than 7% relative to the mode and the mean. Variability in pyroxene is expected, as different pyroxene chemistries have been grouped into only two end-members (pigeonite and augite), a limitation also noted by J. T. S. Cahill et al. (2010). Although assigning a precise degree of accuracy to these values remains challenging, we aim to extend the analysis to a wider range of compositions and chemistries in future tests to improve applicability. For ilmenite, deviations are typically less than 10%, except for sample 15071. In contrast, plagioclase exhibits larger discrepancies, up to 20%–30% in some samples, reflecting the inherent difficulty of resolving weak diagnostic absorption features in plagioclase VSWIR spectra. In contrast with the approach of B. Hapke (2001), our semiempirical framework relies on lab-based measurements of true space-weathered lunar samples, empirically circumventing the need to make assumptions about the distribution of npFe (or microphase iron). Furthermore, it allows us to keep the number of free parameters to half of what would be required under the more

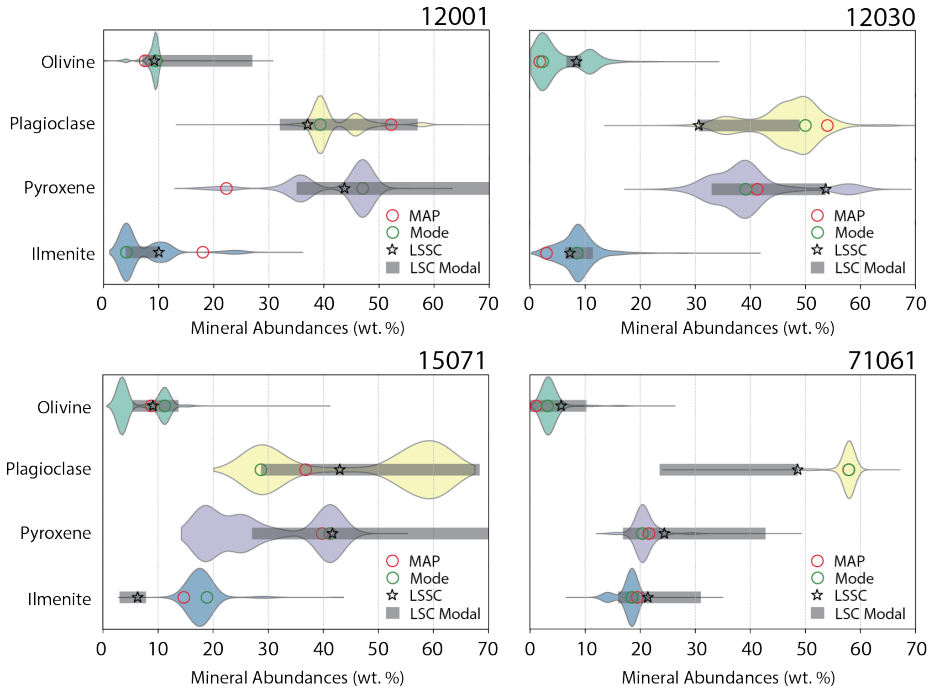



**Figure 4.** Estimated probability densities of mineral abundances for all crystalline phases. Red and green circles indicate MAP and mode estimates, respectively. Stars (Lunar Soil Characterization Consortium, LSCC) and gray bars (multiple studies reported in the represented normalized modal abundances from the Lunar Sample Compendium, LSC) indicate lab-measured compositional measurements, normalized to exclude glasses, agglutinates, and other unspecified minor phases. LSCC pyroxene abundances were reported as "total pyroxene" in LSCC data; accordingly, we show the combined distribution of pigeonite and augite. Plots showing each pyroxene phase separately are provided in Figure F1 of Appendix F.

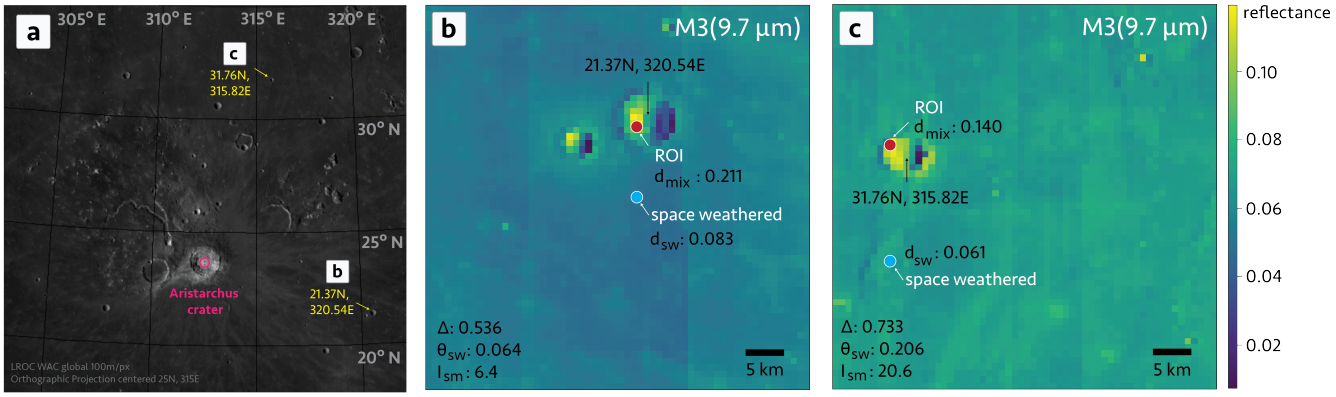



**Figure 5.** Application of space-weathering correction processes to M3 data around fresh craters near the Aristarchus crater. Illustration of the space-weathering correction process: (a) orthographic projection of the Lunar Reconnaissance Orbiter Camera Wide-Angle Camera (LROC WAC) imagery, centered at 25°N, 315°E, showing the Aristarchus crater (pink circle) and regions of interest (ROIs) for spectral weathering correction (yellow arrows). (b), (c) The location of freshly exposed rock (red) and the surrounding space-weathered regolith (blue). The map shows the thermally corrected M3 970 nm reflectance base map.

mechanistic model of B. Hapke (2001), making the inversion of modal mineralogy computationally tractable.

*Application to M3 data.* Relatively shallow, young craters represent ideal targets for our approach because they are more likely to expose relatively fresh materials of similar bulk composition to that of surrounding space-weathered intercrater surfaces. A set of shallow, young craters near Aristarchus was thus selected to illustrate applications of the space-weathering correction method to thermally corrected M3 reflectance spectra for a region that had been previously processed according to the methods of S. Li & R. E. Milliken (2016). Mapped pyroclastic deposits were avoided to ensure that ROIs were dominated by materials similar to the nearby mare-like composition (Figure 5).

First, the thermally corrected M3 reflectance data were converted into SSA (Appendix D). Second, an ROI displaying immature-to-submature characteristics ($I_{sm} < \sim 60$) was selected (red circles in Figure 5(b), (c)). Third, a fully space-weathered end-member spectrum was estimated by assuming that the most space-weathered pixels near the ROI (blue circles in Figure 5(b), (c)) can be used to approximate the fully space-weathered end-member for the ROI. Given the nonlinear relationship between band depths (or $\Delta$) and $I_{sm}$ (Figure 2),

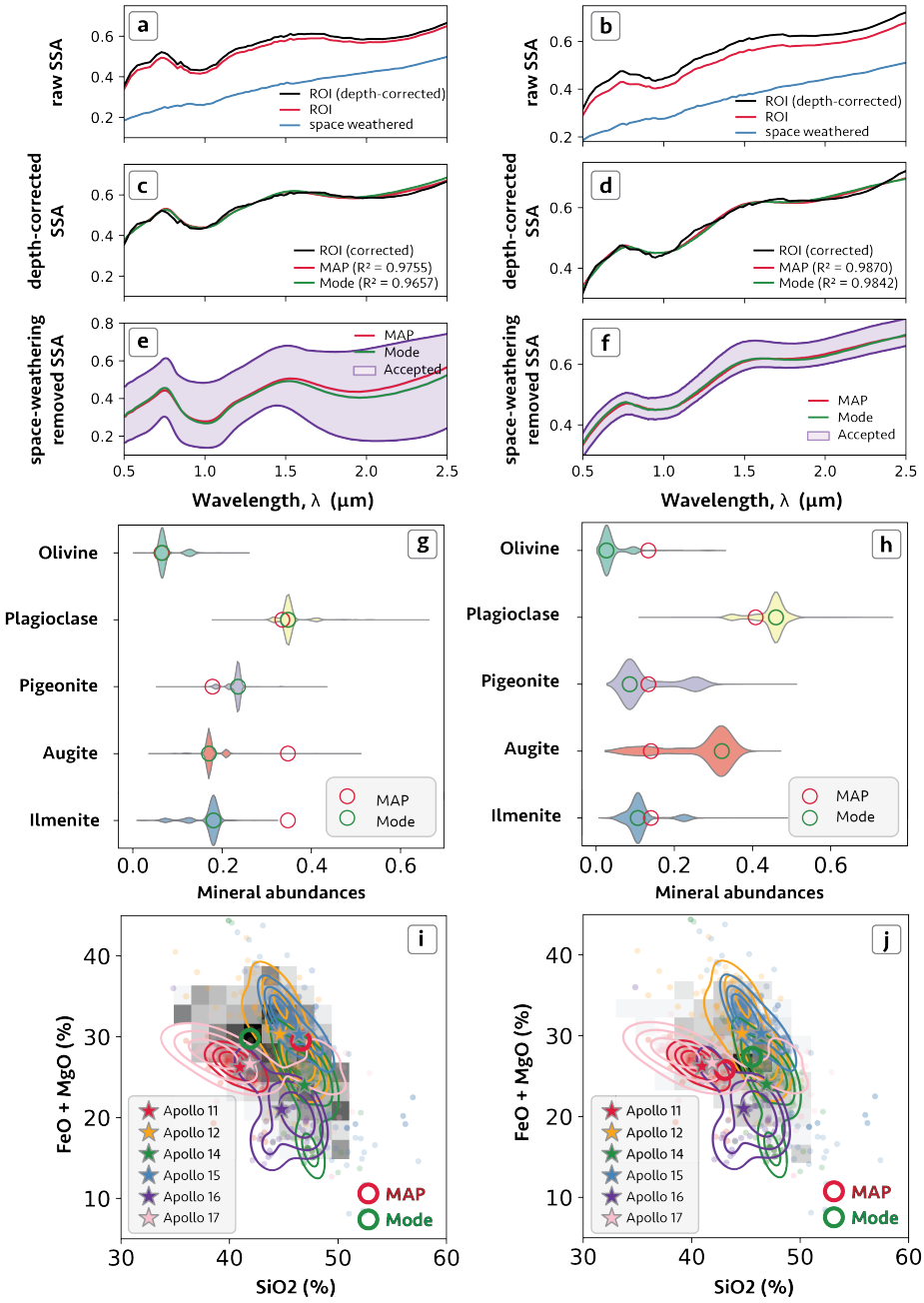



**Figure 6.** Application of space-weathering correction processes to M3 data at the locations shown in Figure 5(b) for panels (a), (c), (e), (g), and (i), and in Figure 5(c) for panels (b), (d), (f), (h), (j). Illustration of the space-weathering correction process: (a), (b) the original SSA from the ROI (red), the SSA of the fully space-weathered end-member (blue), and the depth-corrected SSA (black). (c), (d) The model fits the depth-corrected SSA spectrum (black) from Bayesian spectral unmixing modeling, with MAP (red) and mode (green). (e), (f) Space-weathering corrected SSA spectra (red = MAP, green = mode, and shading = range of accepted models across $10^6$ samples). (g), (h) Posterior probability densities of mineral abundances for all crystalline phases, after exclusion of agglutinates. (i), (j) Estimated FeO + MgO versus $SiO_2$ for all accepted samples (heat map), with MAP (red circle) and mode (green circle) indicated. Transparent dots, stars, and contour plots represent individual laboratory geochemical measurements, averages, and distribution contours from Apollo mare basalt samples, as derived from data in the Apollo lunar sample compendium and cataloged in the ApolloBasalt database (K. A. Cone et al. 2020).

this approximation is justified as long as the selected pixels have $I_{sm} > \sim 60$ and a composition similar to that of the ROI. Finally, the full space-weathering correction workflow was applied to the ROI spectrum using the approximated fully space-weathered end-member (Figure 6). The MCMC algorithm was run with a Markov chain length of $10^6$ using the same compositional end-members as for the LSCC mare basalt soil samples.

Similar to the laboratory samples, rms values for the MAP and mode estimates exceed 0.97, indicating that the "depth correction" yielded intermediate spectra that could be well modeled with the selected nine compositional end-members. The estimated "fresh" lunar soil spectra exhibit deeper band depths, less spectral reddening, and an overall increased albedo, similar to what is observed in fresh SSA spectra of lunar mare basalt soils (Figure 6(a)–(f)). Both groups of

inverted mineral abundances indicate the likely presence of ilmenite, suggesting that the ROI consists of ilmenite mare basalt-type materials. Based on our results, these materials could potentially be classified as Apollo ilmenite basalts in the Lunar Sample and Photo Catalog,[5] which include Apollo 11 low-K and high-K basalts; Apollo 12 ilmenite basalts; and Apollo 17 low-K, high-K, and very high Ti basalts (G. J. Taylor et al. 1991; Figure 6(g), (h)). To facilitate quantitative comparisons with other mare basalt samples gathered from different Apollo landing sites (K. A. Cone et al. 2020), inverted mineral compositions were converted to oxide abundances (heat map in Figure 6(i), (j)). In ($SiO_2$, FeO + MgO) space, inverted mineralogies of these craters significantly overlap with ranges observed at Apollo landing sites, with mode models closely matching mare basalt samples collected during the Apollo 12 and 15 missions.

## 4. Discussion and Conclusions

We developed a framework to correct for the influence of space weathering on VSWIR spectra of the lunar surface using an empirical band depth correction and probabilistic unmixing modeling. This approach was validated against laboratory measurements of Apollo regolith samples and subsequently applied to thermally corrected M3 reflectance data of the lunar surface. The results demonstrate the potential of this workflow to facilitate a computationally tractable, quantitative breakdown of mineralogy for remote compositional analyses on the Moon.

In its current form, our semiempirical framework relies on the assumption that the same band depth correction can be applied broadly to mare basalts. In other words, it assumes that different groups of mare basalts (e.g., illmenite-, olivine-, and pyroxene-rich basalts) follow the same relationship between their band depth at 970 nm and $I_{sm}$ (Equation (3)). As more samples become available, the approach could be refined in future iterations by incorporating different versions of Equation (3) that are tailored to specific basalt types.

Due to the strongly nonlinear relationship between band depths and soil maturity, our method is most effective for areas with immature ($I_{sm}$ = 0–30) or submature soils ($I_{sm}$ = 30–60) rather than mature soils ($I_{sm}$ = 60–100). Using Equation (3), calculated $\Delta$ values for mature soils span a narrow range—$\Delta$ = 0.98–1.00. In contrast, calculated $\Delta$ values in immature and submature soils range from 0.00 to 0.98. In other words, large changes in $I_{sm}$ in mature soils lead to very small changes in band depth, such that band depth is a poor proxy for the degree of space weathering in such samples. This approach, moreover, relies on the availability of a space-weathered target that possesses a mineral composition similar to that of the target region. As a result, shallow young craters that expose relatively fresh materials represent particularly promising targets for this method. This technique could be adapted to other lunar terrain types, such as lunar highlands (notably through the use of different compositional end-members), and possibly serve as a template for space-weathering corrections and remote compositional analyses on other airless bodies.

## Acknowledgments

We thank the editor, Dr. Faith Vilas, for handling our submission, as well as two anonymous reviewers for providing constructive suggestions that improved our manuscript. We also thank Dr. Lori Pigue, who conducted an internal review for the USGS. The codes for space-weathering corrections for this study and formulae related to appendices are available on Zenodo, a permanent CERN-backed online data repository at doi:10.5281/zenodo.17258920 (J. I. Jung et al. 2025).

LSCC samples (C. M. Pieters et al. 2000; L. A. Taylor et al. 2001) are available at https://sites.brown.edu/relab/lscc/ and https://pds-speclib.rsl.wustl.edu/. The USGS spectroscopy library and associated software are hosted at https://www.usgs.gov/labs/spectroscopy-lab/usgs-spectral-library. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

## Appendix A
## SSA of Mare Basalt and Soil Samples

Figure A1 presents the sample spectra used in the main text to derive Equation (3). For the band-depth correction, a reference wavelength of $\lambda$ = 0.97 $\mu$m was adopted.

[5] https://curator.jsc.nasa.gov/lunar/samplecatalog

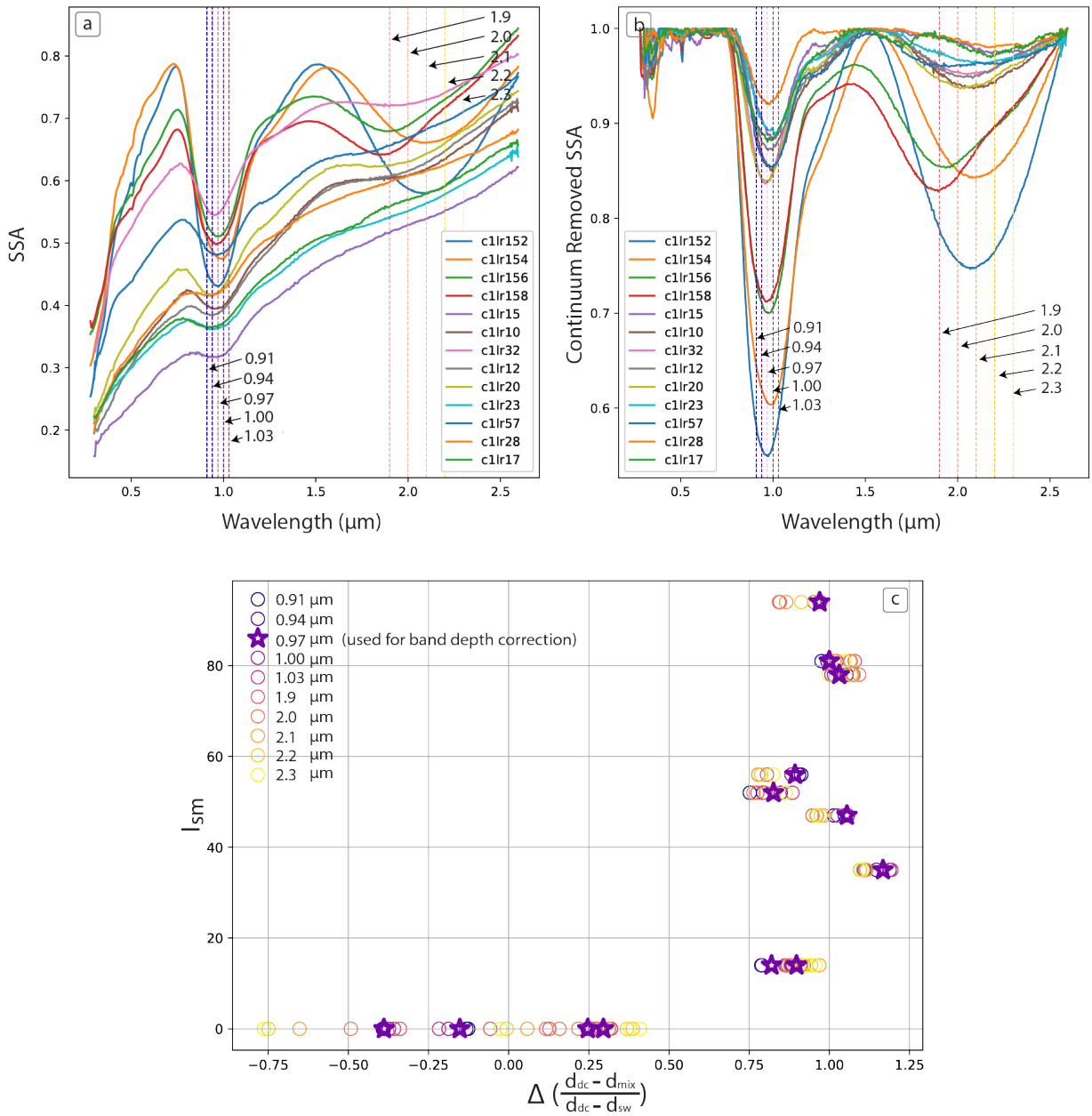



**Figure A1.** (a) Single scattering albedo (SSA) spectra and (b) continuum-removed SSA spectra of mare basalt and soil samples (available at https://sites.brown.edu/relab/lscc/ and https://pds-speclib.rsl.wustl.edu/). Vertical dashed lines indicate the wavelengths selected for band depth correction analysis. (c) Relationship between spectral contrast at various wavelengths and soil maturity, expressed as Ism = Is/FeO (R. V. Morris 1978). The 0.97 $\mu$m band, which is used for band depth correction, is highlighted with an open star marker.

## Appendix B
## MCMC Spectral Unmixing and Prior Distributions

Most spectral compositional analyses to date report a single set of mineral abundances and grain sizes, corresponding to the overall best-fit modeled spectrum (S. Li & L. Li 2011; P. J. Isaacson et al. 2011). However, it is important to recognize that the best-fit model does not necessarily reflect the true composition, due to factors such as noise in the data or slight variations in the chemistries of mineral end-members. Furthermore, trade-offs between mineral abundances and grain sizes create non-unique solutions to the spectral unmixing problem, such that multiple solutions can explain the data equally well (M. G. A. Lapôtre et al. 2017a). To address this issue and provide a fuller picture of sets of possible mineral assemblages and their likelihood, a probabilistic approach was devised by M. G. A. Lapôtre et al. (2017a), utilizing a Bayesian framework applied to the Hapke RTM. This algorithm has been employed to determine the composition of Martian dune sand (M. G. A. Lapôtre et al. 2017b; E. B. Rampe et al. 2018) as well as the surface composition of Ceres (H. Kurokawa et al. 2020).

Applying Bayes's theorem to the probability of a given mineral assemblage (with mineral abundance, $m$, and grain size, $D$) given the SSA data, $d$, yields

$$p(m, D|d) \propto p(d|m, D)p(m, D), \tag{B1}$$

where $p(d|m, D)$ represents the likelihood of a given mineral assemblage given the data and is taken as a multivariate Gaussian distribution of the form

$$p(d|m, D) = \frac{1}{(2\pi^{N_d}|C_\chi|^{\frac{1}{2}})} e^{-\frac{1}{2}[d-G(m,D)]^T C_\chi [d-G(m,D)]}, \quad \text{(B2)}$$

where $C_\chi$ is a covariance matrix that represents the sum of uncertainties due to measurement error and model prediction error and has the same dimension as $d$ (i.e., $2N_d \times 2N_d$, where $N_d$ is the number of wavelength–SSA pairs in the data).

Mineral abundances are drawn from a Dirichlet distribution,

$$p(m) = \mathrm{Dir}(m_1,\ldots,m_N; a_1,\ldots,a_N) = \frac{1}{B(a)} \prod_{i=1}^{N} m_i^{a_i-1}, \quad \text{(B3)}$$

where $N$ is the number of component minerals and $B(a)$ is the multinomial beta function,

$$B(a) = \frac{\prod_{i=1}^{N} \Gamma(a_i)}{\Gamma(\sum_{i=1}^{N} a_i)}, \quad \text{(B4)}$$

with $\Gamma$ the gamma function,

$$\Gamma(a) = \int_0^\infty x^{a_i-1} e^{-x} dx, \quad \text{(B5)}$$

where $a$ is a parameter that changes the shape of the probability distribution. The mean of prior distributions for mineral abundances, $p(D)$, was set to 0.07 (i.e., 7 wt%) for olivine, 0.16 for pigeonite, 0.16 for augite, 0.39 for plagioclase, 0.10 for ilmenite, 0.115 for pyroxene-rich glass, 0.0016 for plagioclase-rich glass, 0.0016 for Apollo orange glass, and 0.0016 for agglutinate, and $\alpha$ was set to 30.

The grain size prior, $p(D)$, was taken as uniform,

$$p(D_i) = \frac{1}{D_{i,\max} - D_{i,\min}} \mathbf{1}_{D_i \in [D_{i,\min}, D_{i,\max}]}, \quad \text{(B6)}$$

where $D_{i,\min}$ and $D_{i,\max}$ are the lower and upper bounds for grain sizes of $i$th mineral. These values were set to 0 and 990 $\mu$m, respectively.

To solve the inverse problem, a set of sample models (up to the length of the Markov chain $L$) is generated by the CATMIP algorithm. The probability of each candidate model is assessed for acceptance or rejection based on the goodness of fit between the synthetic and model SSA spectra. Our sampling approach is described in detail in M. G. A. Lapôtre et al. (2017a), although for this study, we made some minor modifications. CATMIP draws samples of the prior probability density function (PDF) and then simulates a series of intermediate PDFs that transition from the prior to the target posterior PDF. The samples of each intermediate PDF, in turn, are drawn using the Metropolis algorithm. The Metropolis algorithm is a Markov chain (that is, a random walk) that requires the use of a proposal PDF to produce candidate steps for the random walk. To optimize sampling efficiency, for each intermediate PDF, the proposal PDF for abundance values is a Dirichlet distribution fit to the mineral abundance samples of the previous intermediate PDF. Similarly, the proposal PDF for grain sizes is a multivariate normal PDF with covariance calculated from the grain size samples of the previous intermediate PDF.

## Appendix C
## Single Scattering Albedo of a Mixture

The SSA, $\omega_{\mathrm{mix}}$, of an intimate mixture of compositional end-members is calculated as a linear combination of the SSA of individual compositional end-members, i.e.,

$$\omega_{\mathrm{mix}} = \sum_{i=1}^{N} f_i \omega_i, \quad \text{(C1)}$$

where $N$ is the number of mineral end-members and $f_i$ denotes the fractional relative cross-sectional area of the $i$th component given by

$$f_i = \frac{\sigma_i}{\sum_{i=1}^{N} \sigma_i}, \quad \text{(C2)}$$

where

$$\sigma_i = \frac{m_i}{\rho_i D_i}, \quad \text{(C3)}$$

where $m_i$, $\rho_i$, and $D_i$ are the mass abundance, density, and grain size, respectively, of the individual end-member $i$. Here, reflectance spectra from the USGS library and RELAB (Brown University) of olivine, pigeonite, augite, plagioclase, ilmenite, pyroxene-rich glass, plagioclase-rich glass, and agglutinates were used as possible compositional end-members for lunar mare basalt soils (Table C1). The SSA spectra derived using Appendix D are shown in Figure C1.

**Table C1**
Characteristics of Mineral End-members Used

| End-member | $\rho$ (kg m$^{-3}$) | $D$ | $n$ | Sample ID | RELAB Specimen Code | $D_{\mathrm{lab}}$ ($\mu$m) | $D_{\mathrm{used}}$ ($\mu$m) |
|---|---|---|---|---|---|---|---|
| Olivine | 3320 | 0–990 | 1.83 | 15555,965 | LR-CMP-169 | >0, <125 | 80 |
| Pigeonite | 3400 | 0–990 | 1.77 | 70017,535 | LR-CMP-176 | >0, <125 | 80 |
| Augite | 3380 | 0–990 | 1.73 | 70035,188 | LR-CMP-179 | >0, <125 | 80 |
| Plagioclase | 2690 | 0–990 | 1.56 | 70017,535 | LR-CMP-177 | >0, <125 | 80 |
| Ilmenite | 4270 | 0–990 | 2.13 | 70017,535 | LR-CMP-178 | >0, <125 | 80 |
| Px-rich glass | 3300 | 0–990 | 1.64 | synthetic | SC-EAC-053 | >45, <90 | 50 |
| Pl-rich glass | 3000 | 0–990 | 1.64 | synthetic | SC-EAC-050 | >45, <90 | 50 |
| Orange glass | 3000 | 0–990 | 1.64 | 74220,806 | LR-CMP-051 | ⋯ | 150 |
| Agglutinates | 3000 | 0–990 | 1.49 | 79221 | LS-BWD-152 | >125, <250 | 150 |

**Note.** $\rho$ = density; $D$ = grain size range used for Bayesian Hapke model implementation; $D_{\mathrm{lab}}$ = reported grain size range; $D_{\mathrm{used}}$ = grain size used for calculation of imaginary index of refraction; $n$ = assumed real index of refraction.

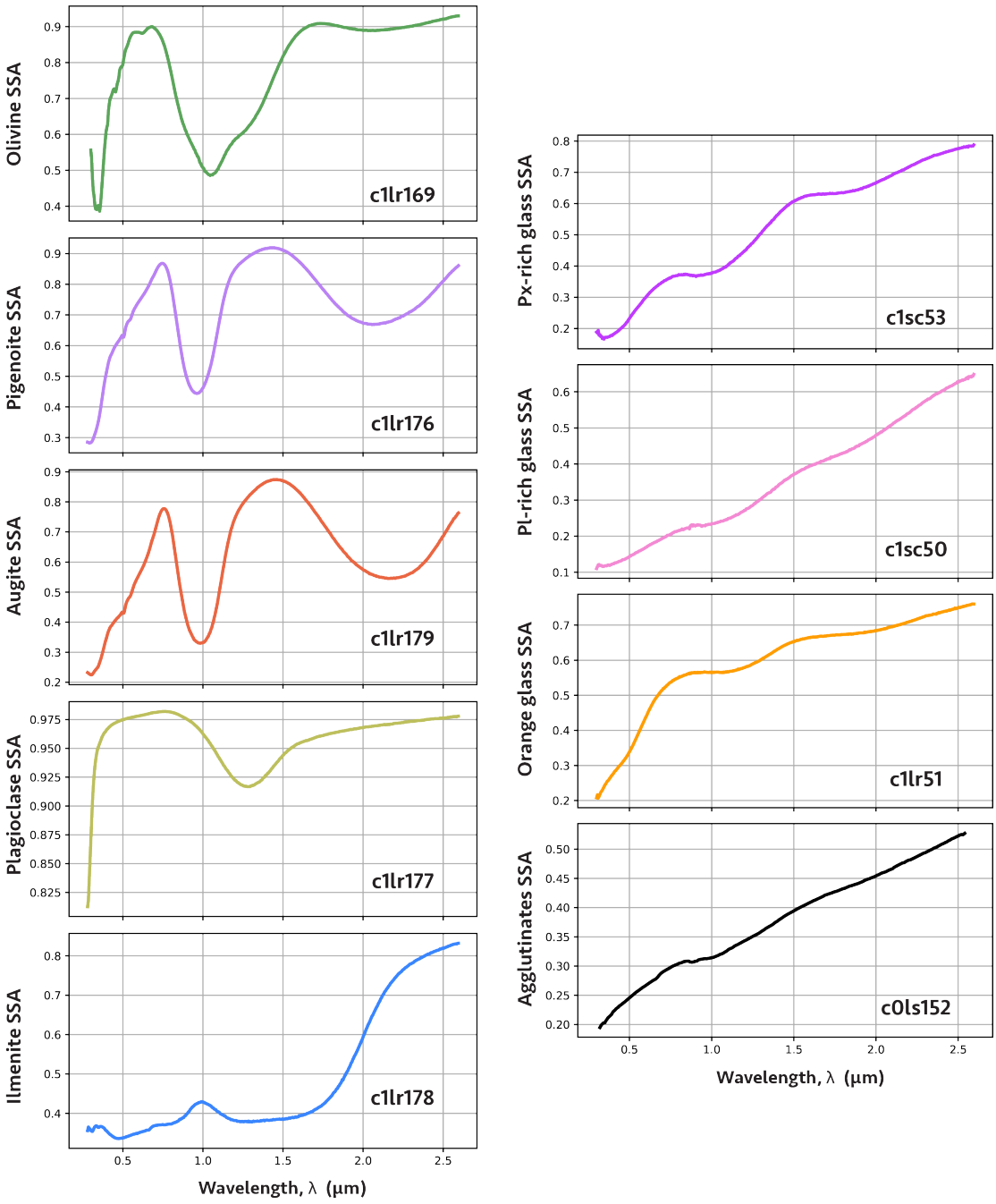



**Figure C1.** Single scattering albedo (SSA) spectra of the nine compositional end-members used in this study.

## Appendix D
## Derivation of Single Scattering Albedo Using the Hapke Radiative Transfer Model

The SSA, $\omega$, is derived from measured reflectance, $r_{\text{measured}}$, using the bisection method to find the root of the function

$$f(\omega) = r_{\text{Hapke}} - r_{\text{measured}}, \tag{D1}$$

where $r_{\text{Hapke}}$ is the radiance coefficient calculated using the Hapke RTM (B. Hapke 1981).

According to the Hapke model,

$$r_{\text{Hapke}} = \frac{\omega}{4(\mu + \mu_0)}[(1 + B(g))P(g) + H(\omega, \mu)\,H(\omega, \mu_0) - 1], \tag{D2}$$

where $\mu$ and $\mu_0$ are the cosines of the incidence emission angles, respectively; $B(g)$ is the backscattering function; $P(g)$ is the phase function; $g$ is the phase angle; and $H$ represents the Chandrasekhar integral function (B. Hapke 1981), approximated

as

$$H(x) = \left[1 - \omega x\left[r_0 + \frac{1 - 2r_o x}{2}\ln\left(\frac{1+x}{x}\right)\right]\right]^{-1}, \quad \text{(D3)}$$

where $r_0 = \frac{1-\gamma}{1+\gamma}$ is the bihemispheric reflectance for isotropic scatters, with $\gamma = \sqrt{1-\omega}$.

Assuming $B = 0$ (no backscattering at the moderate phase angles considered) and $P = 1$ (isotropic scatterer), Hapke's equation for radiance coefficient reduces to

$$r_{\text{Hapke}} = \frac{\omega}{4(\mu + \mu_0)} H(\omega, \mu) H(\omega, \mu_0). \quad \text{(D4)}$$

## Appendix E
## Calculation of Compositional End-member Optical Constants

The complex refractive index of a material as a function of wavelength can be expressed as

$$m(\lambda) = n + ik(\lambda), \quad \text{(E1)}$$

where $n$ (refractive index) and $k$ (extinction coefficient) are the real and imaginary parts of the complex refractory index, $m$, respectively, and are together referred to as the material's optical constants. The imaginary part, $k(\lambda)$, for each compositional end-member was derived using the bisection method to solve for

$$f(k) = r_{\text{Hapke}}(k) - r_{\text{measured}}, \quad \text{(E2)}$$

where $r_{\text{Hapke}}$ is the radiance coefficient calculated using the Hapke RTM as in Appendix D and substituting for SSA:

$$\omega = S_e + (1 - S_e)\frac{1 - S_i}{1 - S_i\theta}\theta, \quad \text{(E3)}$$

where $S_e$ and $S_i$ are the surface reflectance coefficients for externally incident light and internally scattered light, respectively, and $\theta$ is the internal transmission coefficient. $S_e$ and $S_i$ are given by

$$S_e = \frac{(n-1)^2 + k^2}{(n+1)^2 + k^2} + 0.05 \quad \text{(E4)}$$

and

$$S_i = 1.014 - \frac{4}{n(n+1)^2}. \quad \text{(E5)}$$

In turn, $\theta$ can be expressed as

$$\theta = \frac{r_i + \exp(\sqrt{\alpha(\alpha+s)\langle D\rangle})}{1 + r_i\sqrt{\alpha(\alpha+s)\langle D\rangle}}, \quad \text{(E6)}$$

where $r_i$ is the internal diffusive bihemispheric reflectance inside a particle, given as

$$r_i = \frac{1 - \sqrt{\frac{\alpha}{\alpha+S}}}{1 + \sqrt{\frac{\alpha}{\alpha+S}}}\theta, \quad \text{(E7)}$$

where $\alpha$ is the internal absorption coefficient,

$$\alpha = \frac{4\pi k}{\lambda}, \quad \text{(E8)}$$

with $\langle D\rangle$ as the mean-free path of a photon,

$$\langle D\rangle = \frac{2}{3}\left[n^2 + \frac{1}{n^2}(n^2-1)^{\frac{3}{2}}D\right]. \quad \text{(E9)}$$

## Appendix F
## Estimated Probability Density Distributions of Modeled Mineral Abundances

In the main text, inverted pyroxene abundances are reported in combined form (Figure 4). Here, we report the original probability density distributions of modeled mineral abundances, including the two endmember pyroxene phases (pigeonite and augite) as well as olivine, plagioclase, and ilmenite, for four lunar basalt samples (Figure F1).

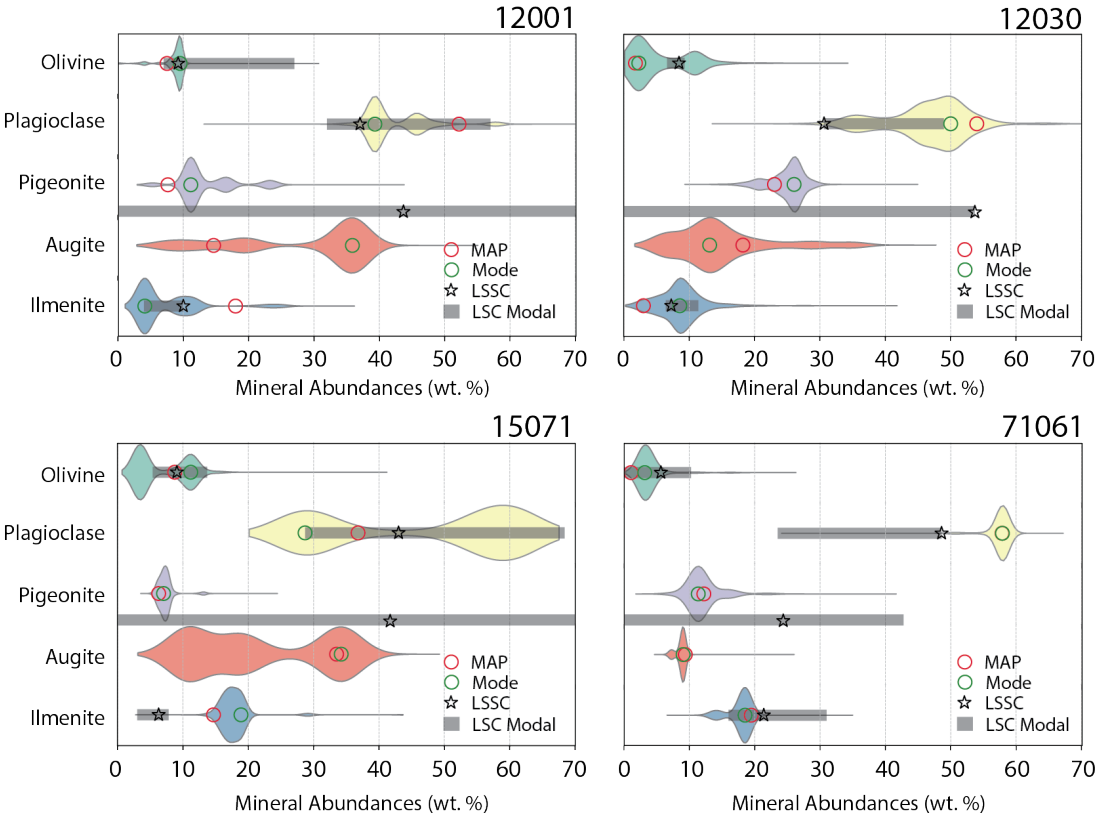



**Figure F1.** Estimated probability density distributions of modeled mineral abundances for olivine, plagioclase, pigeonite, augite, and ilmenite in four lunar basalt samples. These are the same distributions shown in Figure 4, but here, pigeonite and augite are displayed separately rather than combined. Red circles indicate the maximum a posteriori (MAP) estimates, green circles indicate modes of the distributions, black stars represent normalized modal abundances from the Lunar Soil Characterization Consortium (LSCC), and gray bars represent normalized modal abundances from LSCC data and from multiple studies reported in the Lunar Sample Compendium (LSC), renormalized to exclude glass, agglutinates, and other unspecified minor phases. LSCC pyroxene abundances are reported as "total pyroxene" and are thus shown only once, between the pigeonite and augite rows.

## ORCID iDs

Ji-In Jung https://orcid.org/0000-0001-8728-7320

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